The FOMC Announcement Reversal*

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Abstract

This paper documents a strong negative relationship between the pre-FOMC announcement return and the return following the announcement, suggesting that the trend direction of the stock market tends to reverse following the announcement. This reversal phenomenon could be explained by an overreaction to the news before the announcement. For example, as investors slowly digest public and private signals over the previous days, the pre-announcement interval could be characterized by overreaction, likely due to the interaction between newswatchers and momentum traders.

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1 Introduction

The importance of monetary policy announcements has grown exponentially since 1994, when the Federal Open Market Committee (FOMC) began making public announcements of its decisions.[1] Investors closely observe central banks' policy statements, their assessment of the economic outlook, and the information provided about the future direction of monetary policy. As monetary policy decisions exert a considerable influence on asset prices (Gürkaynak et al., 2005), and the stock market reaction is mainly generated by unexpected monetary policy actions (Bernanke and Kuttner, 2005), investors require substantial compensation for bearing the risk associated with scheduled FOMC announcements (Savor and Wilson, 2013; Lucca and Moench, 2015; Cieslak et al., 2019; Brusa et al., 2020). Lucca and Moench (2015) document a large average excess returns in U.S. equities before scheduled FOMC announcements, a phenomenon that they term "pre-FOMC announcement drift."[2] They show that half of the excess return in U.S. equities over the period 1980 – 2011 accrues in the 24 hours before scheduled FOMC announcements, whereas the average post-announcement return is approximately zero. They conclude that this unconditional excess return is not directly related to the actual FOMC monetary policy decision, highlighting a puzzle that is difficult to explain with standard asset pricing theory. Following LM, a vast literature that explores the impact of FOMC announcements on financial markets began studying the pre-FOMC announcement drift and other related patterns around announcements.[3]

[1] Before 1994, there was uncertainty regarding when Fed decisions would reach financial markets (Thornton, 2005). The day that the Federal Reserve releases the policy statement from a regularly scheduled FOMC meeting is defined henceforth as "event," "FOMC announcement day," or "FOMC day."

[2] In the remainder of this article, we refer to "Lucca and Moench (2015)" as "LM," and to "pre-FOMC announcement drift" as "Drift", representing the average pattern of the pre-FOMC announcement returns or the strategy consisting of buying the asset the day prior the announcement and selling it just before the announcement. Similarly, we refer to "post-FOMC announcement return" as "post-announcement return." The same applies to "pre-FOMC announcement return." All times mentioned in this paper are in Eastern Standard Time (EST).

[3] Potential explanations for the Drift range from the possibility of information leakage before FOMC announcements (Bernile et al., 2016; Ai and Bansal, 2018; Ying, 2020; Mano, 2021) and informal communication by Fed officials with the media and the financial sector (Cieslak et al., 2019; Cieslak and Schrimpf, 2019; Vissing-Jorgensen, 2019; Cieslak and Vissing-Jorgensen, 2021) to differences of opinion (Cocoma, 2018), constrained institutional buying pressure (Kaul and Watanabe, 2015), endogenous information acquisition (Ai et al., 2021), a premium for heightened uncertainty (Hu et al., 2022), and rare disasters (Wachter and Zhu, 2022).

This paper revisits the impact of FOMC announcements on returns by studying the relationship between pre- and post-announcement return on scheduled FOMC meetings. In contrast with the previous literature, which focuses on the average performance of the pre- or post-announcement return, we account for the heterogeneity in pre-announcement returns and we show that there is a strong negative relationship between the pre-announcement return and the return following the announcement, suggesting that the trend direction of the stock market tends to reverse following the announcement.

While the previous literature shows that the post-announcement return is, on average, close to zero (Lucca and Moench, 2015; Hu et al., 2022), it did not explore the time series variation of the post-announcement return and how it is related with the pre-announcement return. Only a handful of papers find interesting patterns related to the stock market behaviour from the monetary policy announcement (or just before) until the end of the trading day, and the main findings are related to the impact of contractionary and expantionary monetary policy shocks. For instance, Rigobon and Sack (2004) and Gürkaynak et al. (2005) find that an unanticipated 25 basis points increase in the short-term interest rate results in decline in the S&P 500 Index of over 1%. Similarly, Indriawan et al. (2021) document large post-announcement returns occur mostly following expansionary policy shocks, while contractionary policy shocks yield negative returns, and Ozdagli and Weber (2017) find a larger effect of surprise monetary easing on stock returns than of surprise tightening.[4]

First, using intraday data on E-Mini S&P 500 futures, we document a strong negative relationship between pre-announcement return and post-announcement return on FOMC days, suggesting that FOMC meetings characterized by a positive pre-announcement return are usually characterized by a negative post-announcement return, and vice versa for meetings characterized by a negative pre-announcement return. This negative relationship, which is reported in Figure 1, also holds when accounting for the magnitude of the pre-announcement return.

INSERT FIGURE 1

[4] Given the well-known negative correlation between stock returns and changes in volatility (e.g., Black, 1976; Christie, 1982; French et al., 1987; Campbell and Hentschel, 1992; Bekaert and Wu, 2000; Wu, 2001), a positive post-announcement return would be consistent with the decrease in investor fear after the announcement, due to the resolution of uncertainty, and the associated downward volatility jumps (Fernandez-Perez et al., 2017; Amengual and Xiu, 2018; Beckmeyer et al., 2020). According to Hu et al. (2022), the post-announcement return-to-variance ratio is far exceeded by the pre-announcement counterpart, suggesting that distinct types of risk premia are realized around the announcement window.

Second, to explore the magnitude of the FOMC announcement reversal, we propose a reversal strategy (henceforth, Reversal), which consists of buying (selling) E-Mini S&P 500 at 1:50 p.m. EST of scheduled FOMC days, if the pre-announcement return is negative (positive), and closing the position two hours later. We test Reversal and we compare it with trading strategies that investors implement around FOMC announcements. During the period 1997 – 2020, considering 180 scheduled FOMC announcements, Reversal generates an average return of about 18 basis points (bps) per event, net of transaction costs, and an annualized Sharpe ratio of 0.56. In a subsample analysis, we distinguish between the (Quasi) Lucca and Moench period (1997 – 2011) and the post-Lucca and Moench period (2011 – 2020), and we document that the difference in the Sharpe ratio of Reversal and Drift narrows in the first period, whereas in the second period Drift disappears and Reversal generates greater Sharpe ratios than before.

Third, to explore whether the reversal phenomenon is driven by specific variables, we condition Reversal on the (ex-ante) level of uncertainty and credit conditions, and we study how it relates to monetary policy and macroeconomics news. Conditioning Reversal on different measures of uncertainty and credit conditions, we find that Reversal is not driven by their level, as it persists when uncertainty is low and credit conditions are loose. Conditioning Reversal on the type of monetary policy news of the current announcement, we find that it generates positive Sharpe ratios independent of the type of monetary policy news, and remarkable Sharpe ratios in announcements characterized by a positive Federal funds rate shock and in meetings associated with the Summary of Economic Projections. A reversal strategy that incorporates the monetary policy news of the announcement, by opening the position 20 minutes after the announcement, still performs well. In this case, the performance is lower than if an investor opens the position before the announcement, and it is mostly driven by unexpected announcements. Conditioning Reversal on (recent) macroeconomics news, we find that it performs particularly well in FOMC announcements that follow a recent negative CPI surprise, a negative initial jobless claim surprise, a negative surprise in new home sales, and/or a positive surprise in ISM manufacturing.

Our results are robust to different definitions of the reversal strategy and of the pre- and post-announcement return. We obtain similar results when implementing Reversal over different time intervals: instead of opening the position at 1:50 p.m. and closing it exactly 2 hours later, we compute Reversal by opening the position once for every minute from 1:51 p.m. until 1:59 p.m. and closing the position exactly 2 hours thereafter each time. We confirm the negative relationship between pre- and post-announcement returns on scheduled FOMC meetings

by computing the pre-announcement return starting from the FOMC day opening time, instead of using the 24-hour pre-announcement return. Basing the reversal strategy on the direction of the pre-announcement return, computed from the FOMC day opening time until 1:50 p.m., instead of from 1:50 p.m. $FOMC_{t-1}$, we find that the strategy generates similar returns and Sharpe ratios. In contrast, when computing Reversal by closing the position 24 hours after it was opened, we find that the return of the strategy is similar to the 2-hour Reversal, suggesting that price reversal occurs mainly until the closing of the FOMC day.

Our conjecture is that these results could be explained by an overreaction to the news before the announcement. Investors (newswatchers) may slowly digest public and private signals over the days prior to the announcement. This slow adjustment, according to Hong and Stein (1999), may interact with the presence of momentum traders, leading to overreaction. In support of this conjecture, we show that the pre-announcement interval is characterized by momentum, while there is no evidence of momentum over similar intervals on non-FOMC days. This conjecture is related to the literature on informal communication by Fed officials with the media and the financial sector (Cieslak et al., 2019; Cieslak and Schrimpf, 2019; Vissing-Jorgensen, 2019; Cieslak and Vissing-Jorgensen, 2021), according to which informal communication from the Federal Reserve has had a large stock market impact. This pattern could also arise due to the occurrence of leakages prior to the announcement, as discussed by Bernile et al. (2016); Ai and Bansal (2018); Ying (2020); Mano (2021).

The remainder of this paper proceeds as follows. Section 2 describes the data. The reversal phenomenon is illustrated in Section 3. Section 4 explores the magnitude of the FOMC announcement reversal and whether this phenomenon is driven by specific variables. Robustness tests are presented in Section 5. Conclusions are offered in Section 6.

2 Data Description

Our study relies primarily on intraday data of the bid and ask price of E-Mini S&P 500 futures (continuous contract).[5] We use data at a one-minute frequency of the closing value of every minute from September 1997, when E-Mini S&P 500 was launched, until the end of January 2020 (the last week of the last scheduled FOMC meeting before the COVID-19

[5] E-Mini S&P 500 futures are the primary futures trading vehicle for the S&P 500 Index. For detailed information, see `http://www.cmegroup.com/trading/equity-index/us-index/e-mini-sandp500_contract_specifications.html`.

outbreak began being incorporated into monetary policy decisions). We use E-Mini S&P 500 futures, as they are characterized by deep liquidity and tight bid-ask spreads and are one of the most efficient, liquid, and cost-effective ways to gain market exposure to the S&P 500 Index.

Over the time period of the empirical analysis, which spans from September 1997 to January 2020, there were a total of 180 scheduled FOMC meetings, eight regularly scheduled meetings per year. From the website of the Federal Reserve, we obtained data and information on FOMC meetings, the Federal funds rate, and monetary policy decisions. Immediately after the conclusion of every scheduled meeting, the Fed releases a policy statement.[6] Policy statements were released, on average, at approximately 2:15 p.m. EST until March 14, 2011; we obtained the exact release time from the internet appendix of LM (Table IA.I). Since March 2011, policy statements have been released at 2:00 p.m. EST, except for eight meetings when the policy statement was released at 12:30 p.m. EST.[7]

The main variables of interest in this paper are the FOMC pre- and post-announcement return. We compute the pre-announcement return as the return of the mid-price of E-Mini S&P 500 from 24 hours and 15 minutes before the announcement to 15 minutes before the announcement, in line with LM. We distinguish between announcements characterized by a positive and a negative pre-announcement return, and we define announcements as characterized by a large (small) positive pre-announcement return if the positive pre-announcement return is higher (lower) than its median, where the median is computed accounting only for positive pre-announcement returns. Similarly, we define announcements as characterized by a large (small) negative pre-announcement return, if the negative pre-announcement return is lower (higher) than its median, where the median is computed accounting only for negative pre-announcement returns. The post-announcement return is the return computed from the mid-price opening value of the minute of the announcement until 3:59 p.m. EST.

As a proxy for realized volatility, we use the difference between the high and low value of the minute. We use $Volatility_{Morning,t\text{-}1}$, which we compute as the average volatility across minutes of the morning interval, as an alternative control variable to VIX. The morning interval is set to start at the opening time and extends until 24 hours and 15 minutes before the FOMC

[6] The FOMC started announcing the outcomes of meetings in February 1994. In February 1995, the FOMC announced that all monetary policy changes would be communicated to the public immediately. In January 2000, the FOMC announced that, following each regularly scheduled meeting, it would issue a statement even if there had been no change in monetary policy.

[7] For information on FOMC meetings and monetary policy decisions, refer to `https://www.federalreserve.gov/monetarypolicy/fomccalendars.htm`.

announcement for $\text{FOMC}_{\text{t-1}}$ days. For non-$\text{FOMC}_{\text{t-1}}$ days, the morning interval is set to start at the opening time and ends at 2:00 p.m. (1:45 p.m.) EST until (after) March 14, 2011.

Information on macroeconomics news, such as the release date and the actual and survey value of each macroeconomic variable, is obtained from Bloomberg Terminal.

Information on uncertainty and credit conditions indicators, which we use to better understand the reversal phenomenon, is reported in the Appendix.

3 Reversal Around FOMC announcements

In this section, we first present the reversal phenomenon by analyzing the time series relationship between pre- and post-announcement return on scheduled FOMC meetings, accounting for the magnitude of the pre-announcement return. Then, we address potential explanations for Reversal in the equity market, from a theoretical perspective.

3.1 The FOMC Announcement Reversal

Unlike the previous literature, which focuses on the average behavior of the stock market in the interval before or after the announcement, this paper focuses on the time series variation in both pre- and post-announcement return on scheduled FOMC meetings. To test the relationship between pre- and post-announcement return, we run the following regressions:

$$\begin{aligned} \textit{Post-Ann. Return}_t = \alpha + \beta_1 \times \textit{Pre-Ann}_t \times \textit{FOMC}_t + \beta_2 \times \textit{Pre-Ann}_t \times \textit{Non-FOMC}_t \\ + \beta_3 \times VIX_{\textit{Previous Week}} + \epsilon_t \end{aligned} \quad (1)$$

$$\textit{Post-Ann. Return}_t = \alpha + \beta_1 \times \textit{Pre-Ann}_t \times \textit{Positive Pre-Ann}_t + \beta_2 \times VIX_{\textit{Previous Week}} + \epsilon_t \quad (2)$$

$$\begin{aligned} \textit{Post-Ann. Return}_t = \alpha + \beta_1 \times \textit{Pre-Ann}_t \times \textit{Positive Pre-Ann}_{\textit{Large},t} \\ + \beta_2 \times \textit{Pre-Ann}_t \times \textit{Positive Pre-Ann}_{\textit{Small},t} + \beta_3 \times VIX_{\textit{Previous Week}} + \epsilon_t \end{aligned} \quad (3)$$

where, for FOMC days, the dependent variable is the post-announcement return, while $\textit{Pre-Ann}_t$ indicates the pre-announcement return. For non-FOMC days, they are a proxy for FOMC pre- and post-announcement return: the former is computed starting at 2:00 p.m. (1:45 p.m.) of the previous day until (after) March 14, 2011, while the latter starts 15 minutes after the former and extends until the end of the trading day. $\textit{FOMC}_t$ is a dummy variable equal to 1 when the trading day is characterized by a scheduled FOMC announcement, while $\textit{Non-FOMC}_t$ is a

dummy variable equal to 1 when the trading day is not an FOMC day (also excludes $\text{FOMC}_{\text{t-1}}$ and $\text{FOMC}_{\text{t+1}}$). $Positive\ Pre\text{-}Ann_t$ and $Positive\ Pre\text{-}Ann_{Large,t}$ ($Positive\ Pre\text{-}Ann_{Small,t}$) are dummy variables equal to 1 when the pre-announcement return is positive or large (small) positive. Equations (2) and (3) are also estimated using a dummy for negative, instead of positive, pre-announcement return. $VIX_{Previous\,Week}$ represents the VIX level seven trading days before each observation. Equation (1) is estimated using daily data over the period 09/10/1997 – 01/31/2020, while the other equations are estimated using data on our 180 events only.

Table 1 reports estimates of equations (1) to (3). Column (1) shows that there is a negative relationship between pre- and post-announcement return on FOMC days only, as the coefficient of the interaction term with the non-FOMC dummy is not statistically significant. A 1% return in the pre-announcement interval implies about 18 bps of negative return in the post-announcement interval. Columns (2) and (4) show that this relationship is also negative when we account, separately, for the direction of the pre-announcement return (positive or negative). Columns (3) and (5) show that there is a negative and statistically significant relationship when the pre-announcement return is largely positive or largely negative; when the pre-announcement return is small, the point estimates still suggest a negative relationship, although not statistically significant. The results from Table 1 are robust to controlling for realized volatility instead of VIX.

INSERT TABLE 1

This negative relationship is shown in Figure 2, which reports the average intraday E-Mini S&P 500 cumulative return across FOMC meetings, using one observation every 5 minutes. Figure 2 distinguishes between whether the pre-announcement return is positive or negative, as well as if it is high or low, according to the value of the 24-hour return ending 15 minutes before the announcement. Figure 2 suggests that there is a reversal after the announcement, independent of the magnitude and direction of the pre-announcement return.

INSERT FIGURE 2

In Figure 3, which reports the average E-Mini S&P 500 cumulative return starting at 1:45 p.m. EST on FOMC days, we focus on the post-announcement return to see whether price reversal is stronger when the pre-announcement return is positive or negative. We start at 1:45 p.m. FOMC_{t}, as it represents the end of the pre-announcement period for events with announcement at 2:00 p.m. (the earliest announcement time in our sample, excluding the eight

announcements at 12:30 p.m.). As shown in Figure 3, going long E-Mini S&P 500 when the pre-announcement return is negative yields, on average, a greater return than going short E-Mini S&P 500 when the pre-announcement return is positive. In contrast, the average unconditional post-announcement return is close to zero.

INSERT FIGURE 3

3.2 Momentum in the Pre-Announcement Period

In this paragraph, we address potential explanations for the FOMC announcement reversal phenomenon from a theoretical perspective. The empirical results presented in the previous paragraph could be explained by an overreaction to the news before the announcement. Newswatchers may slowly digest public and private signals over the days prior to the announcement. This slow adjustment may interact with the presence of momentum traders, leading to overreaction, in line with previous findings from Hong and Stein (1999). This conjecture is related to the literature on informal communication by Fed officials with the media and the financial sector (Cieslak et al., 2019; Cieslak and Schrimpf, 2019; Vissing-Jorgensen, 2019; Cieslak and Vissing-Jorgensen, 2021), according to which informal communication from the Federal Reserve has had a large stock market impact. This pattern could also arise due to the occurrence of leakages prior to the announcement, as discussed by Bernile et al. (2016); Ai and Bansal (2018); Ying (2020); Mano (2021).

To explore the presence of momentum before FOMC announcements, we divide the 24-hour pre-announcement interval into three returns: the return from 24 hours and 15 minutes before the announcement until the closing time of $\text{FOMC}_{\text{t-1}}$ ($Return_{Afternoon,t-1}$); the return from the closing time of $\text{FOMC}_{\text{t-1}}$ until 15 minutes before the announcement ($Return_{Close,t-1:Ann.}$); and the return from the opening time of the FOMC day until 15 minutes before the announcement ($Return_{Open,t:Ann.}$). We want to test whether the first return predicts the second and the third return, as, in this case, the pre-announcement interval would be characterized by momentum. For this purpose, we run the following regressions:

$$Return_{Close,t-1:Ann.} = \alpha + \beta_1 \times Return_{Afternoon,t-1} \times FOMC_t + \beta_2 \times Return_{Afternoon,t-1} \times Non\text{-}FOMC_t + \beta_3 \times VIX_{Previous\,Week} + \epsilon_t \quad (4)$$

$$Return_{Open,t:Ann.} = \alpha + \beta_1 \times Return_{Afternoon,t-1} \times FOMC_t + \beta_2 \times Return_{Afternoon,t-1} \times Non\text{-}FOMC_t + \beta_3 \times VIX_{Previous\,Week} + \epsilon_t \quad (5)$$

where, for non-FOMC days, $Return_{Afternoon,t-1}$ is computed starting at 2:00 p.m. (1:45 p.m.) of the previous day until (after) March 14, 2011 and extends until the end of the same trading day, while the other two returns extend until 24 hours after the start of $Return_{Afternoon,t-1}$. $FOMC_t$ is a dummy variable equal to 1 when the trading day is characterized by a scheduled FOMC announcement, while $Non\text{-}FOMC_t$ is a dummy variable equal to 1 when the trading day is not an FOMC day (also excludes $\text{FOMC}_{\text{t-1}}$ and $\text{FOMC}_{\text{t+1}}$). $VIX_{Previous\,Week}$ represents the VIX level seven trading days before each observation. The two equations are estimated using daily data over the period 09/10/1997 – 01/31/2020.

The results are reported in Table 2, which shows that there is evidence of momentum in the 24-hour pre-announcement interval, as, on FOMC days, there is a positive and statistically significant relationship between $Return_{Afternoon,t-1}$ and the two subsequent returns. In contrast, there is no statistical evidence for non-FOMC days, which are thus characterized neither by reversal nor by momentum. As for the case of Table 1, the results from Table 2 are robust to controlling for realized volatility instead of VIX.

INSERT TABLE 2

4 The Magnitude of the FOMC Announcement Reversal

In this Section, to explore the magnitude of the FOMC announcement reversal, we introduce and test a reversal strategy (Reversal). We start by exploring the magnitude of Reversal and comparing it with Drift and with other patterns around the announcement. Then, to explore whether this phenomenon is driven by specific variables, we condition Reversal on the level of uncertainty and credit conditions, and we study how it relates to monetary policy and macroeconomics news.

4.1 Comparing Reversal with other Patterns Around the Announcement

As documented in the previous Section, there is a strong negative relationship between pre- and post-announcement return on scheduled FOMC meetings, independent of the magnitude and direction of the pre-announcement return. This negative relationship suggests that FOMC meetings characterized by a positive pre-announcement return are, on average, characterized by a negative post-announcement return, and vice versa for meetings characterized by a negative

pre-announcement return.[8].

In this paragraph, we test a reversal strategy, which consists of buying (selling) E-Mini S&P 500 if the pre-announcement return is negative (positive) and closing the position two hours later. In particular, Reversal is implemented as follows: if the prior 24-hour E-Mini S&P 500 return at 1:50 p.m. FOMC_t is negative (positive), we buy (sell) E-Mini S&P 500 at 1:50 p.m. FOMC_t and sell (buy) at 3:50 p.m. FOMC_t. We assume that an investor is buying at the ask price and selling at the bid price, considering the closing value of the minute.[9] In Section 5, we implement robustness tests using different time intervals.

Along with Reversal, we test also Drift and other trading strategies that an investor can implement around FOMC announcements. We compute the average return of each strategy across events and test whether it is greater (when positive) or lower (when negative) than zero by implementing a t-test. The Sharpe ratio is computed dividing the average return of the strategy across events by the standard deviation of the strategy returns and is annualized multiplying by $\sqrt{8}$ (since there are 8 FOMC meetings per year). The performance of the strategies is net of transaction costs, as we compute returns by buying at the ask price and selling at the bid price. We ignore brokerage fees, which are close to zero for trading E-Mini S&P 500, and potential market impacts.

We begin by implementing Reversal over the period 1997 – 2020. As shown in Panel A of Table 3, Reversal generates an average return of 17.86 bps per event and an annualized Sharpe ratio of 0.56. In contrast, Drift, which reflects the performance over a 24-hour period (instead of just 2 hours), generates higher returns and Sharpe ratios. Nearly half of the difference in return between Drift and Reversal is due to one extreme event (October 29, 2008), when Drift was greater than 10%; without this extreme observation, the average return of Drift over the full sample period would be 25.53 bps (instead of 31.10 bps). A trading strategy that combines

[8] This pattern is not correlated with the liquidity premium pattern associated with the post-announcement return, as highlighted by Baglioni et al. (2022)

[9] Our measure of pre-announcement return is similar to the pre-FOMC announcement drift of LM, which is the 24-hour return ending at 2:00 p.m. EST on the day of scheduled FOMC announcements. In their sample period, announcements are at approximately 2:15 p.m., while in our sample period, the average announcement time is at approximately 2:05 p.m., and 1:50 p.m. is a good approximation of 15 minutes before the average announcement time. For the eight meetings when the policy statement is released at 12:30 p.m. EST, the pre-announcement return is computed from 12:15 p.m. $\text{FOMC}_{\text{t-1}}$ to 12:15 p.m. FOMC_t; in this case, Reversal is implemented by opening the position 15 minutes before the announcement and closing the position 1 hour and 45 minutes after the announcement.

Drift and Reversal is able to return nearly 50 bps over a 26-hour period, on average, and a Sharpe ratio greater than that of Drift. In accordance with Lucca and Moench (2015) and Hu et al. (2022), we find that the post-announcement return is very close to zero: it is slightly negative, on average, and not statistically significant. In Panel B of Table 3, we examine the return of Reversal, considering the magnitude and direction of the pre-announcement return. Reversal is stronger when the pre-announcement return is negative, particularly when the pre-announcement return is highly negative. In some cases, we find no statistical evidence of the average (positive) return of the strategy being greater than zero: a plausible explanation is that we are testing a small sample of extreme realizations.

INSERT TABLE 3

4.2 Reversal, Uncertainty, and Credit Conditions

As previous literature document that the pre-FOMC announcement drift occurs mainly in periods of considerable market uncertainty (Martello and Ribeiro, 2018; Neuhierl and Weber, 2019), we investigate the magnitude of Reversal in accordance with the current level of uncertainty. For this purpose, we restrict the sample by conditioning Reversal to the (ex-ante) high and low levels of uncertainty, using the measures of uncertainty described in the Appendix.

The results are reported in Table 4, which shows that the Sharpe ratio of Reversal is positive across all measures of uncertainty, independent of the level of uncertainty being high or low, and the return is statistically greater than zero in most cases. When looking at the difference in the average performance of Reversal when uncertainty is high versus low, we find that Reversal tends to be much more pronounced for the case of high market uncertainty and high monetary policy uncertainty, although it does not disappear when the level of uncertainty is low and the Sharpe ratios remain relevant.

INSERT TABLE 4

As shown in Table 4, results are similar when we condition Reversal to credit conditions: when credit spreads and credit conditions are tight, Reversal tends to be more pronounced, but it does not disappear when credit spreads narrow and credit conditions are loose.

4.3 Reversal and Monetary Policy News

We now study how Reversal is related to monetary policy news. First, we study the relationship between Reversal and the monetary policy news of the current and previous FOMC

meetings. Second, we study whether it might be profitable to implement a reversal strategy starting after the announcement, and if we can improve the performance by conditioning the strategy on both the policy news revealed in the announcement and the direction of the pre-announcement return.

4.3.1 Reversal Without Knowing the Outcome of the Announcement

Nakamura and Steinsson (2018) construct monetary shocks using unexpected changes in interest rates over a 30-minute window surrounding scheduled Federal Reserve announcements, from 10 minutes before the announcement to 20 minutes after. In particular, they construct Federal funds rate shocks (unexpected changes in the Federal funds rate) as a 30-minute change in the expectations of the Federal funds rate, whereas they construct policy news shocks as the first principal component of the unanticipated change over the 30-minute windows in the following five interest rates: the Federal funds rate immediately following the FOMC meeting, the expected Federal funds rate immediately following the next FOMC meeting, and expected three-month Eurodollar interest rates at horizons of two, three, and four quarters. We identify Federal funds rate and policy news shocks using data from both Nakamura and Steinsson (2018) and Acosta (2022), which updated previous data from Nakamura and Steinsson (2018).[10]

We define expected monetary policy announcements as announcements characterized by no Federal funds rate shock (no 30-minute change in the expectations of the Federal funds rate immediately after the announcement). In contrast, unexpected monetary policy announcements are defined as announcements characterized by a Federal funds rate shock, regardless of whether the shock is positive or negative. We also distinguish among announcements characterized by a positive or a negative Federal funds rate shock; a positive or a negative policy news shock (there are no announcements characterized by no policy news shock); the Summary of Economic Projections (SEP); and unconventional monetary policy announcements (forward guidance and balance sheet announcements).

Table 5 shows how Reversal performs in accordance with different types of monetary policy news. Again, the strategy is implemented assuming that an investor enters its long or short position before every FOMC announcement, at 1:50 p.m. EST, thus being unaware of the

[10] In particular, we use data from Nakamura and Steinsson (2018), available at `https://eml.berkeley.edu/~enakamura/papers.html`, from the September 1997 meeting until the March 2014 meeting. We use data from Acosta (2022), available at `https://www.acostamiguel.com/research`, from the April 2014 meeting until the January 2020 meeting.

monetary policy news. Overall, the Sharpe ratio of Reversal is positive, regardless of the type of monetary policy news. Reversal performs particularly well in meetings associated with the SEP. This finding is in line with Gu et al. (2018), who document sizable positive unconditional stock returns after FOMC announcements accompanied by the SEP and press conferences, which are characterized by greater uncertainty resolution than other FOMC meetings. Bodilsen et al. (2021) document that excess stock returns are strongly and positively related to their betas on announcement days with press conferences. We find that meetings associated with the SEP are characterized by a small negative Drift, followed by a positive post-announcement return, on average. Reversal also performs well when the monetary policy announcement is unexpected, results driven by the performance of the strategy in meetings characterized by a positive Federal funds rate shock, and when there is a positive policy news shock and/or a forward guidance announcement. In the latter case, there is no statistical evidence of the average return of the strategy being greater than zero, as we are testing a small sample of extreme realizations.

INSERT TABLE 5

Table 6 shows how Reversal performs given the monetary policy news of the previous FOMC meeting. Again, Reversal generates positive Sharpe ratios overall, particularly when the previous meeting is characterized by a positive policy news shock; a positive Federal funds rate shock; an expected announcement; and/or an unconventional announcement. Reversal performs well also when the previous announcement is not characterized by a hike in the Federal funds rate.

INSERT TABLE 6

4.3.2 Reversal Knowing the Outcome of the Announcement

As shown in Figure 3, the equity market tends to be quite stable, on average, in a short window around the announcement. Therefore, we now ask whether we can improve the performance of Reversal by starting the strategy after the announcement and conditioning it on both the monetary policy news revealed in the announcement and on the magnitude of the pre-announcement return. As Nakamura and Steinsson (2018) construct monetary shocks using unexpected changes in interest rates over a 30-minute window surrounding the announcement, from 10 minutes before the announcement to 20 minutes after, this time the reversal strategy starts 20 minutes after each announcement and ends at 3:50 p.m.

The results are reported in Table 7. In Panel A, we replicate Table 6 with the exclusion of SEP and unconventional announcements, which are characterized by a small number of observations. The average performance of the reversal strategy that starts 20 minutes after the announcement is almost 50% lower than the performance of Reversal from Table 3, and it is mainly driven by unexpected announcements. Conditioning the strategy on Federal funds rate or policy news shocks, we find that the return is lower than in Table 3 and it is slightly negative, even though not statistically significant, when there is a negative Federal funds rate shock. These results suggest that the stock market moves a lot in a short window after the announcement, but a relevant part of the return remains unexploited 20 minutes after the announcement.

INSERT TABLE 7

In Panel B, we combine the information on the magnitude of the monetary policy shock of the current meeting with the magnitude of the pre-announcement return. As we expect a positive (negative) post-announcement return following a negative (positive) monetary policy shock, and a positive (negative) post-announcement return following a negative (positive) pre-announcement return, we combine monetary policy shocks and pre-announcement returns of the same sign (positive or negative). The average performance of the strategy is significantly improved for the case of a negative Federal funds rate shock and of a negative policy news shock.

4.4 Reversal and Macroeconomics News

In this paragraph, we study how Reversal relates to macroeconomics news. First, we implement Reversal conditioning the strategy on recent macroeconomics news. Second, we study the performance of a reversal strategy on trading days characterized by a macroeconomic announcement.

4.4.1 Reversal Conditioned on Recent Macroeconomics News

In order to study the performance of Reversal in accordance with recent macroeconomics news, we select the following macroeconomics variables: Consumer confidence; CPI; Initial jobless claims; ISM manufacturing; New home sales; and Retail sales. For each variable, we distinguish whether, on each release date, it is characterized by a positive surprise, a negative surprise, or by no surprise. We define a positive (negative) surprise, when the actual value of

the macroeconomic news is greater (lower) than the survey value. When the actual value of the macroeconomic news is equal to the survey value, we refer to "no surprise."[11] Only CPI is characterized by no surprise a significant number of times. In our sample, macroeconomic news released on the same day of an FOMC announcement is always released prior to the FOMC announcement.

In Table 8, we report the performance of Reversal given the surprise of the latest macroeconomics news. On FOMC days, Reversal tends to work regardless of the recent macroeconomics news surprise. In fact, the average return of the strategy is always positive and (most of the times) statistically significant. Reversal performs particularly well in FOMC announcements that follow a recent negative CPI surprise, a negative initial jobless claim surprise, a negative surprise in new home sales, and/or a positive surprise in ISM manufacturing. We obtain similar results by conditioning the reversal strategy to the surprise of the following macroeconomics news: Durable goods orders; Housing starts; Industrial production; ISM services; Nonfarm payrolls; Sentiment.

INSERT TABLE 8

4.4.2 Reversal and Macroeconomic Announcements

Our paper is also related to the broader literature on the macroeconomic announcement premium. Savor and Wilson (2013) document a significant equity market return in the U.S. on days with major macroeconomic announcements, and Brusa et al. (2020) show that the same holds for many other countries. Kurov et al. (2019) examine E-Mini S&P 500 futures and Treasury futures around the release time of 30 U.S. macroeconomic announcements and find that 9 announcements are characterized by the presence of a drift, which might originate from a combination of information leakage and superior forecasting that incorporates proprietary data. Bernile et al. (2016) find no evidence of informed trading during lockups ahead of nonfarm payrolls, U.S. Producer Price Index, and GDP data releases. Hu et al. (2022) find large overnight returns with no abnormal variance before nonfarm payrolls, ISM, and GDP announcements, similar to the pre-FOMC returns. Wang (2015) and Illeditsch et al. (2021)

[11] Note that a positive (negative) surprise does not imply that the macroeconomic news has been positive (negative) for the economy. For instance, a positive CPI surprise implies that the latest CPI release has been higher than expected, thus being a negative news for the economy. Vice versa, a positive consumer confidence reading implies an improvement of consumer's perception of the economy.

find that stock market returns in the U.S. and abroad tend to underreact to macroeconomics news.

We now ask whether Reversal is a phenomenon that characterizes FOMC announcements only, or, as for the case of Drift, if it is also present in some macroeconomic announcements. We implement Reversal with the 30 U.S. macroeconomic announcements examined by Kurov et al. (2019), which they grouped into the following eight categories: Income (GDP advance; GDP preliminary; GDP final; Personal income); Employment (ADP employment; Initial jobless claims; Nonfarm employment); Industrial activity (Factory orders; Industrial production); Investment (Construction spending; Durable goods orders; Wholesale inventories); Consumption (Advance retail sales; Consumer credit; Personal consumption); Housing sector (Building permits; Existing home sales; Housing starts; New home sales; Pending home sales); Government (Government budget); Net exports (Trade balance); Inflation (Consumer Price Index; Producer Price Index); and Forward-looking indices (Conference Board Consumer Confidence Index; Index of Leading Indicators; ISM Manufacturing Index; ISM Non-Manufacturing Index; University of Michigan CSI preliminary; University of Michigan CSI final).[12] We compute Drift and Reversal over different time intervals but find no strong evidence of a reversal following macroeconomic announcements.[13] In particular, we separately compute 1- and 2-hour Reversal returns and Sharpe ratios given the direction of the pre-announcement return over the 24 hours and over the 1 hour period ending 15 minutes before the macroeconomic announcement. The return of the strategy is always lower than 10 basis points, except for the case of trading days characterized by the release of nonfarm payrolls and trading days characterized by the release of GDP (advanced estimate). While Reversal generates positive returns and Sharpe ratios for more than half of the macroeconomic announcements that we consider, an average Reversal return lower than 10 basis points does not compensate, in our opinion, for the risk that an investor would bear to trade around FOMC or macroeconomic announcements.

We thus document that Reversal is a peculiarity of FOMC announcements, as this strategy is able to generate consistent returns and Sharpe ratios mainly on FOMC days. Interestingly, Reversal generates slightly negative returns following the release of FOMC minutes.

[12] For further details and information regarding the frequency and the release times of these announcements, refer to Table 1 of Kurov et al. (2019).

[13] The results for Reversal associated with macroeconomic announcements are available upon request.

5 Robustness Tests

In this section, we test the robustness of our results. First, we implement Reversal over two different periods: the (Quasi) Lucca and Moench period and the Post-Lucca and Moench period. Second, we modify the reversal strategy by implementing it over different time intervals and by adopting different definitions of the pre- and post-announcement return.

5.1 Subsample Analysis

5.1.1 Reversal in the (Quasi) Lucca and Moench Period

Previous studies provide evidence that Drift is particularly relevant in the Lucca and Moench period, while in the subsequent period, Drift appears to disappear (Kurov and Gu, 2016). Boguth et al. (2019) extend the sample to September 2017 and find that the Drift is limited to announcements with press conferences. Extending the sample to 2019, Kurov et al. (2021) show that the Drift has not only disappeared for announcements without press conferences but also weakened for announcements with press conferences. Reduced uncertainty is a possible explanation that they give for this change. Similarly, Ben Dor and Rosa (2019) find no Drift from 2011 to 2017, while Neuhierl and Weber (2019) find a large and positive Drift for a sample from 1997 until 2002 and then again for a sample from 2007 until 2009, before the Drift levels off until 2019. Guo et al. (2021) find that the stock market increases significantly over the pre-FOMC announcement window only during periods of high investor sentiment and low economic policy uncertainty, while Martello and Ribeiro (2018) and Neuhierl and Weber (2019) document that the Drift occurs mainly in periods of considerable market uncertainty. In contrast, Baglioni and Ribeiro (2023) show that the ex-ante level of the corporate bond market distress is a good predictor for the Drift, subsuming the relevant information of equity market uncertainty. Neuhierl and Weber (2021) find that U.S. stock returns begin drifting up 25 days before expansionary monetary policy surprises, whereas they decrease before contractionary surprises.

We now implement Reversal and other possible strategies that an investor can implement around FOMC announcements, restricting the sample to what we call the (Quasi) Lucca and Moench period, which spans from September 1997 to March 2011. The LM period spans from 1994 to 2011, but we have data only from 1997, when E-Mini S&P 500 started trading. The results are reported in Table 9. Panel A documents that the difference in the Sharpe ratio between Drift and Reversal narrows in the (Quasi) LM period, in comparison to the full

sample period reported Table 3, although Reversal continues to outperform. Compared to the full sample period, Drift is much stronger, while Reversal is relatively weaker and the post-announcement return remains slightly negative and not significantly lower than zero. Excluding October 29, 2008, when Drift was 10.29%, the average return of Drift over the (Quasi) LM period would be approximately 18% lower (41.25 bps instead of 50.31 bps), suggesting that a relevant part of the pre-FOMC announcement drift puzzle highlighted by LM is due to one extreme event. Panel B shows that Reversal works for every kind of pre-announcement return (on average, generates positive returns) and is stronger in the case of a high negative pre-announcement return. Reversal in the (Quasi) LM period is shown in Figure 4.

INSERT TABLE 9

INSERT FIGURE 4

5.1.2 Reversal in the Post-Lucca and Moench Period

We now implement Reversal and other possible strategies that an investor can implement around FOMC announcements, restricting the sample to what we call the post-Lucca and Moench period, which spans from April 2011 to January 2020. The results are reported in Table 10. Panel A documents that, in the post-LM period, Reversal generates very high Sharpe ratios, while, as expected, Drift disappears. Both the post-announcement return and the open-to-close FOMC return are slightly negative in the post-LM period, while the close $\text{FOMC}_{\text{t-1}}$ to close FOMC_{t} return and the return from one day before the announcement to close FOMC_{t} are slightly positive, although none of them is statistically significant. Only the return of Reversal and the combination of Drift and Reversal are positive and statistically significant. Panel B shows that Reversal generates strong Sharpe ratios for every kind of pre-announcement return, particularly for the case of a high negative pre-announcement return. Reversal in the post-LM period is shown in Figure 5.

INSERT TABLE 10

INSERT FIGURE 5

5.2 Modifying the Reversal Strategy

We test the robustness of our results by implementing the reversal strategy over different time intervals and by adopting different definitions of the pre- and post-announcement return.[14]

First, we implement Reversal over different time intervals. Instead of opening the position at 1:50 p.m. and closing it exactly 2 hours afterwards, we compute Reversal by opening the position once for every minute from 1:51 p.m. until 1:59 p.m. and closing the position exactly 2 hours afterwards each time. We find that returns and Sharpe ratios are similar to those reported in Table 3, in line with Figure 3, which shows that the stock market tends to be stable in a short window before the announcement and close to the end of the trading day, on average.

Second, we compute the pre-announcement return from the FOMC day opening time (9:30 a.m.) until one minute before the announcement (instead of using the 24-hour pre-announcement return). We confirm the negative relationship between pre-announcement and post-announcement returns on FOMC days, as in Table 1, and we also find statistical evidence for the positive relationship between these returns when computed over the same time interval for non-FOMC days, suggesting that Reversal is a characteristic of FOMC days only. Buying E-Mini S&P 500 at 9:30 a.m. FOMC_t and closing the position at 1:50 p.m. FOMC_t yields an average return of 16.73 basis points and an annualized Sharpe ratio of 0.96. Then, we implement Reversal given the magnitude and direction of the pre-announcement return, which is now computed from the FOMC day opening time until 1:50 p.m., instead of using the 24-hour pre-announcement return that ends at 1:50 p.m. We re-compute Table 3 to Table 8, and we find that Reversal performs as well as using the magnitude and direction of the 24-hour pre-announcement return as a reference to open the position.

Third, we explore the relationship between pre- and post-announcement returns using a 24-hour post-announcement return, and we find no statistical significance, although the point estimates remain negative. Implementing Reversal by closing the position 24 hours after it was opened, instead of 2 hours afterwards, an investor would obtain a lower annualized Sharpe ratio (0.37 versus 0.56), and a return very close to that of the 2-hour Reversal (17.75 bps versus 17.86 bps), suggesting that price reversal after FOMC announcements happens mainly until the closing of the FOMC day. The return and Sharpe ratio of the 24-hour Reversal remains positive regardless of the level and the measure of uncertainty.

[14] The results of the robustness tests described in this paragraph are available upon request.

6 Conclusions

This paper revisits the patterns in asset returns around FOMC announcements by studying the relationship between pre- and post-announcement returns around scheduled FOMC meetings. We document a strong negative relationship between pre- and post-announcement return on FOMC days, suggesting that the trend direction of the stock market tends to reverse following the announcement. This phenomenon could be explained by an overreaction to the news before the announcement, with investors slowly digesting public and private signals over the previous days. As we show that the pre-announcement interval is characterized by momentum, the interaction between newswatchers and momentum traders may lead to overreaction, in the spirit of Hong and Stein (1999).

To explore the magnitude of the FOMC announcement reversal, we propose a reversal strategy consisting of buying (selling) E-Mini S&P 500 before the announcement if the pre-announcement return is negative (positive) and closing the position two hours later. Over the period 1997 – 2020, this strategy generates remarkable returns and Sharpe ratios. Dividing the full sample period into the (Quasi) LM period and the post-LM period, we find evidence of Reversal in both periods. The average performance of Reversal is positive, independent of uncertainty, credit conditions, monetary policy news, and macroeconomics news. These results are robust to different definitions of the reversal strategy and of the pre- and post-announcement return.

The results of this paper are of interest and importance to policymakers and investors, as this paper sheds the light on (previously) unknown equity return patterns around scheduled FOMC decisions and improves the understanding of financial markets functioning during periods of uncertainty. While this paper is about equity return dynamics around FOMC announcements, we explore whether there is evidence of Reversal in the foreign exchange market as well. In accordance with Mueller et al. (2017), we find evidence of Drift in advance of scheduled FOMC announcements for the euro-dollar exchange rate, but we do not find any evidence of Reversal following the announcement.

7 Appendix - Uncertainty and Credit Indicators

As a proxy for market uncertainty, we use daily data on the VIX closing price, obtained from Bloomberg Terminal. As a proxy for economic and monetary policy uncertainty, we use the Economic Policy Uncertainty (EPU) Index (Baker et al., 2016) and the Monetary Policy

Uncertainty (MPU) Index (Husted et al., 2020), respectively.[15] For every FOMC day, we define the following uncertainty indicators:

I. High (Low) Market Uncertainty: when, on the last trading day of the $FOMC_{t-1}$ week, the VIX closing value is greater (not greater) than its previous median, which is computed using data on the last trading day of previous $FOMC_{t-1}$ weeks only from January 1990 until the previous observation.

II. High (Low) Economic Policy Uncertainty: when the EPU Index of the FOMC day is greater (not greater) than its previous median, which is computed using data on FOMC days only from February 1985 until the previous FOMC day.

III. High (Low) Monetary Policy Uncertainty: when the MPU Index of the FOMC day is greater (not greater) than its previous median, which is computed using data on FOMC days only from February 1985 until the previous FOMC day.

We use two variables that reflect credit conditions: credit spreads and the Chicago Fed National Financial Conditions Credit Subindex (NFCI Credit).[16] For every FOMC day, we define the following credit indicators:

I. Tight (Narrow) Credit Spreads: when, on the last trading day of the $FOMC_{t-1}$ week, the ICE BofA Option-Adjusted Spreads value is greater (not greater) than its previous median, which is computed using daily data from December 1996 until the previous observation.

[15] Both the EPU Index and the MPU Index are constructed by tracking the frequency of newspaper articles related to policy uncertainty. The EPU Index reflects the frequency of articles in 10 leading U.S. newspapers that contain the following trio of terms: "economic" or "economy"; "uncertain" or "uncertainty" and one or more of "Congress," "deficit," "Federal Reserve," "legislation," "regulation," or "White House." The MPU Index is computed by searching for articles in the Washington Post, Wall Street Journal, and New York Times containing the triplet of "uncertainty" or "uncertain"; "monetary policy(ies)" or "interest rate(s)" or "Federal fund(s) rate" or "Fed fund(s) rate"; and "Federal Reserve" or "the Fed" or "Federal Open Market Committee" or "FOMC". Data on the EPU Index is available at `https://fred.stlouisfed.org/series/USEPUINDXD`, while data on the MPU Index is available at `https://sites.google.com/site/lucasfhusted/data`.

[16] As a measure of credit spreads, we use the ICE BofA U.S. High Yield Index Option-Adjusted Spread (OAS), which reflects the spread between a computed OAS index of all bonds rated BB or below (U.S. dollar denominated below investment grade rated corporate debt publicly issued in the U.S. domestic market) and a spot Treasury curve. The NFCI Credit provides a comprehensive weekly update on U.S. credit conditions. Data on the ICE BofA U.S. High Yield Index OAS is available at `https://fred.stlouisfed.org/series/BAMLH0A0HYM2`, while data on the NFCI Credit is available at `https://fred.stlouisfed.org/series/NFCICREDIT`.

II. Tight (Loose) Credit Conditions: when NFCI Credit is positive (negative) on the week before the FOMC announcement, as positive (negative) values of NFCI Credit indicate credit conditions that are tighter (looser) than average.

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8 Tables

Table 1: The Relationship Between Pre- and Post-Announcement Returns.

	Dependent variable:				
	Post-Ann. Return				
	(1)	(2)	(3)	(4)	(5)
Constant	-1e-03**	-2e-03	-1e-03	-2e-04	-1e-03
	(4e-04)	(3e-03)	(3e-03)	(3e-03)	(3e-03)
Pre-Ann × FOMC	-0.178***				
	(0.062)				
Pre-Ann × Non-FOMC	-0.0001				
	(0.020)				
Pre-Ann × Positive Pre-Ann		-0.205*			
		(0.108)			
Pre-Ann × Large Positive Pre-Ann			-0.207*		
			(0.109)		
Pre-Ann × Small Positive Pre-Ann			-0.627		
			(0.526)		
Pre-Ann × Negative Pre-Ann				-0.398**	
				(0.194)	
Pre-Ann × Large Negative Pre-Ann					-0.394**
					(0.195)
Pre-Ann × Small Negative Pre-Ann					-1.610
					(1.233)
$VIX_{\text{Previous Week}}$	1e-04**	1e-04	1e-04	-5e-05	-3e-05
	(2e-05)	(2e-04)	(2e-04)	(1e-04)	(2e-04)
FOMC Days Only	*X*	✓	✓	✓	✓
R^2	0.009	0.026	0.029	0.027	0.032

Note: This table reports the relationship between pre- and post-announcement returns. For FOMC days, the dependent variable is the post-announcement return, while "Pre-Ann" indicates the pre-announcement return. For non-FOMC days, they are a proxy for FOMC pre- and post-announcement returns. "FOMC" is a dummy variable equal to 1 when the trading day is characterized by a scheduled FOMC announcement, while "Non-FOMC" is a dummy variable equal to 1 when the trading day is not an FOMC day (also excludes $FOMC_{t-1}$ and $FOMC_{t+1}$). "Positive (Negative) Pre-Ann," "Large Positive (Negative) Pre-Ann," and "Small Positive (Negative) Pre-Ann" are dummy variables equal to 1 in accordance with the magnitude and direction of the pre-announcement return. "$VIX_{\text{Previous Week}}$" represents the VIX level seven trading days before each observation. The equation in column (1) is estimated using daily data over the period 09/10/1997 – 01/31/2020, while the other equations are estimated using daily data on our 180 events only. Robust standard errors are reported in parentheses. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Table 2: Momentum Before FOMC Announcements.

	Dependent variable:	
	$\text{Return}_{\text{Close,t-1:Ann.}}$	$\text{Return}_{\text{Open,t:Ann.}}$
	(1)	(2)
Constant	1e-03	2e-04
	(5e-04)	(4e-04)
$\text{Return}_{\text{Afternoon,t-1}} \times$ FOMC	0.130^{**}	0.173^{***}
	(0.063)	(0.049)
$\text{Return}_{\text{Afternoon,t-1}} \times$ Non-FOMC	0.068	0.050
	(0.051)	(0.037)
$\text{VIX}_{\text{Previous Week}}$	-3e-05	-1e-05
	(3e-05)	(2e-05)
FOMC Days Only	*X*	*X*
R^2	0.003	0.003

Note: This table explores the presence of momentum in the pre-announcement interval. The 24-hour pre-announcement interval is divided into three returns: the return from 24 hours and 15 minutes before the announcement until the closing time of $\text{FOMC}_{\text{t-1}}$ ($\text{Return}_{\text{Afternoon,t-1}}$); the return from the closing time of $\text{FOMC}_{\text{t-1}}$ until 15 minutes before the announcement ($\text{Return}_{\text{Close,t-1:Ann.}}$); and the return from the opening time of the FOMC day until 15 minutes before the announcement ($\text{Return}_{\text{Open,t:Ann.}}$). For non-FOMC days, the three returns are computed over similar time intervals. "FOMC" is a dummy variable equal to 1 when the trading day is characterized by a scheduled FOMC announcement, while "Non-FOMC" is a dummy variable equal to 1 when the trading day is not an FOMC day (also excludes $\text{FOMC}_{\text{t-1}}$ and $\text{FOMC}_{\text{t+1}}$). "$\text{VIX}_{\text{Previous Week}}$" represents the VIX level seven trading days before each observation. The two equations are estimated using daily data over the period 09/10/1997 – 01/31/2020. Robust standard errors are reported in parentheses. Statistical significance at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Table 3: Reversal and Other Trading Strategies Around FOMC Announcements.

Panel A – Different Strategies

Strategy	Holding Period	Events	Return	Sharpe Ratio
Reversal	From 1:50 p.m. $FOMC_t$ to 3:50 p.m. $FOMC_t$	180	17.86***	0.56
Drift	From 1:50 p.m. $FOMC_{t-1}$ to 1:50 p.m. $FOMC_t$	180	31.10***	0.78
Drift + Reversal	From 1:50 p.m. $FOMC_{t-1}$ to 3:50 p.m. $FOMC_t$	180	48.97***	1.02
Announcement Return	From the Ann. Minute to 3:50 p.m. $FOMC_t$	180	-4.35	-0.13
Open to Close $FOMC_t$	From 9:30 a.m. $FOMC_t$ to 3:50 p.m. $FOMC_t$	180	17.54**	0.49
Close to Close $FOMC_t$	From 3:50 p.m. $FOMC_{t-1}$ to 3:50 p.m. $FOMC_t$	180	22.68***	0.59

Panel B – Reversal Strategies

Reversal, Given:	Strategy	Events	Return	Sharpe Ratio
Negative Pre-Ann.	Buy at 1:50 p.m. and Sell at 3:50 p.m. $FOMC_t$	66	22.92**	0.80
Positive Pre-Ann.	Sell at 1:50 p.m. and Buy at 3:50 p.m. $FOMC_t$	114	14.93**	0.45
Large Negative Pre-Ann.	Buy at 1:50 p.m. and Sell at 3:50 p.m. $FOMC_t$	33	32.69**	1.05
Large Positive Pre-Ann.	Sell at 1:50 p.m. and Buy at 3:50 p.m. $FOMC_t$	57	12.29	0.33
Small Negative Pre-Ann.	Buy at 1:50 p.m. and Sell at 3:50 p.m. $FOMC_t$	33	13.15	0.51
Small Positive Pre-Ann.	Sell at 1:50 p.m. and Buy at 3:50 p.m. $FOMC_t$	57	17.58*	0.60

Note: This table reports the results of "Reversal" over the period 1997 – 2020. "Reversal" is implemented as follows: if the prior 24-hour return at 1:50 p.m. $FOMC_t$ is negative (positive), buy (sell) at 1:50 p.m. and sell (buy) at 3:50 p.m. In Panel A, we compare "Reversal" with other possible strategies that an investor can implement around FOMC announcements, where the position is always opened with a buy order. In Panel B, we condition "Reversal" on the magnitude and direction of the prior 24-hour pre-announcement return (Pre-Ann.). Each strategy is implemented buying at the ask price and selling at the bid price, considering the closing value of each minute (except for the announcement minute, where the open value of the minute is used). "Events" indicates the number of FOMC meetings included. "Return" represents the average E-Mini S&P 500 return of the strategy across FOMC meetings, expressed in basis points. The Sharpe ratio is computed dividing the average return of the strategy (across events) by the standard deviation of the strategy returns and is annualized. Statistical significance of "Return" at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Table 4: Reversal Conditioned on Uncertainty and Credit.

Uncertainty and Credit Measures	Level	Events	Reversal	Difference	Sharpe Ratio
Market Uncertainty	High	83	28.19**	19.16*	0.72
	Low	97	9.03*		0.39
Economic Policy Uncertainty	High	87	21.39**	6.83	0.58
	Low	93	14.56**		0.56
Monetary Policy Uncertainty	High	113	24.71***	18.41*	0.81
	Low	67	6.30		0.21
Credit Spreads	Tight	96	21.87**	8.59	0.59
	Narrow	84	13.28**		0.56
Credit Conditions	Tight	69	23.87**	9.74	0.58
	Loose	111	14.13**		0.59

Note: This table reports the performance of "Reversal" in accordance with the level of uncertainty or credit over the period 1997 – 2020. The measures of uncertainty and credit are described in the Appendix. Each strategy is implemented buying at the ask price and selling at the bid price, considering the closing value of each minute. "Events" indicates the number of FOMC meetings included. "Reversal" represents the average E-Mini S&P 500 return of the strategy across FOMC meetings, expressed in basis points. "Difference" reports the difference in the average performance of "Reversal" when uncertainty (credit) is high (tight) versus low (narrow or loose). The Sharpe ratio is computed dividing the average return of the strategy (across events) by the standard deviation of the strategy returns and is annualized. Statistical significance of "Reversal" at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Table 5: Reversal and Monetary Policy News of the Current Meeting.

FOMC Meeting Characterized By:	Events	Reversal	Sharpe Ratio
Expected Announcement	58	5.76	0.27
Unexpected Announcement	122	23.61***	0.67
Positive Fed Funds Rate Shock	52	48.36***	1.26
Negative Fed Funds Rate Shock	70	5.22	0.16
Positive Policy News Shock	113	22.75***	0.82
Negative Policy News Shock	67	9.61	0.26
Summary of Economic Projections	36	40.73***	1.51
Forward Guidance Announcement	21	25.65	0.64
Balance Sheet Policies Announcement	20	2.46	0.08

Note: This table reports the performance of "Reversal" in accordance with different types of monetary policy news of the current FOMC meeting over the period 1997 – 2020. Each strategy is implemented buying at the ask price and selling at the bid price, considering the closing value of each minute. "Events" indicates the number of FOMC meetings included. "Reversal" represents the average E-Mini S&P 500 return of the strategy across FOMC meetings, expressed in basis points. The Sharpe ratio is computed dividing the average return of the strategy (across events) by the standard deviation of the strategy returns and is annualized. Statistical significance of "Reversal" at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Table 6: Reversal and Monetary Policy News of the Previous Meeting.

Previous Meeting Characterized By:	Events	Reversal	Sharpe Ratio
Expected Announcement	59	22.40**	0.86
Unexpected Announcement	121	15.65**	0.46
Positive Fed Funds Rate Shock	51	29.26**	0.79
Negative Fed Funds Rate Shock	70	5.73	0.18
Positive Policy News Shock	113	27.35***	0.86
Negative Policy News Shock	67	1.85	0.06
Summary of Economic Projections	36	11.38	0.55
Forward Guidance Announcement	21	19.40	0.68
Balance Sheet Policies Announcement	20	23.28**	1.14
Rate Hike	32	0.49	0.02
Rate Cut	22	28.24	0.61
No Rate Change	126	20.46***	0.67

Note: This table reports the performance of "Reversal" in accordance with different types of monetary policy news of the previous FOMC meeting over the period 1997 – 2020. Each strategy is implemented buying at the ask price and selling at the bid price, considering the closing value of each minute. "Events" indicates the number of FOMC meetings included. "Reversal" represents the average E-Mini S&P 500 return of the strategy across FOMC meetings, expressed in basis points. The Sharpe ratio is computed dividing the average return of the strategy (across events) by the standard deviation of the strategy returns and is annualized. Statistical significance of "Reversal" at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Table 7: Reversal Knowing the Outcome of the Announcement.

Panel A – Opening the Position After the Announcement

FOMC Meeting Characterized By:	Events	Reversal	Sharpe Ratio
Reversal	180	9.34*	0.34
Expected Announcement	58	1.17	0.07
Unexpected Announcement	122	13.22*	0.42
Positive Fed Funds Rate Shock	52	31.86***	0.95
Negative Fed Funds Rate Shock	70	-0.63	-0.02
Positive Policy News Shock	113	12.50**	0.51
Negative Policy News Shock	67	4.01	0.12

Panel B – Combining Policy Shocks with Pre-Ann. Return

FOMC Meeting Characterized By:	Events	Reversal	Sharpe Ratio
Positive FFR Shock & Positive Pre-Ann.	32	27.78**	0.80
Negative FFR Shock & Negative Pre-Ann.	24	13.91	0.56
Positive PN Shock & Positive Pre-Ann.	72	10.56*	0.43
Negative PN Shock & Negative Pre-Ann.	25	8.09	0.32
Positive FFR and PN Shocks & Positive Pre-Ann.	26	19.45	0.63
Negative FFR and PN Shocks & Negative Pre-Ann.	13	27.47	0.86

Note: This table reports the performance of a reversal strategy implemented opening the position 20 minutes after each announcement over the period 1997 – 2020. In Panel A, we condition the reversal strategy to different types of monetary policy shocks of the current FOMC meeting. In Panel B, we combine the pre-announcement return (Pre-Ann.) with the Federal Funds Rate (FFR) and/or the Policy News (PN) shock of the same sign (either positive or negative). Each strategy is implemented buying at the ask price and selling at the bid price, considering the closing value of each minute. "Events" indicates the number of FOMC meetings included. "Reversal" represents the average E-Mini S&P 500 return of the strategy across FOMC meetings, expressed in basis points. The Sharpe ratio is computed dividing the average return of the strategy (across events) by the standard deviation of the strategy returns and is annualized. Statistical significance of "Reversal" at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Table 8: Reversal Conditioned on the Latest Macro News.

Macroeconomics News	Surprise	Events	Reversal	Sharpe Ratio
Consumer Confidence	Positive	99	11.46*	0.40
Consumer Confidence	Negative	78	22.10**	0.64
CPI	Positive	47	1.03	0.03
CPI	Negative	61	26.95**	0.85
CPI	No Surprise	72	21.14**	0.66
Initial Jobless Claims	Positive	83	8.08	0.25
Initial Jobless Claims	Negative	94	25.78***	0.84
ISM Manufacturing	Positive	85	18.94**	0.67
ISM Manufacturing	Negative	90	15.44*	0.46
New Home Sales	Positive	80	8.12	0.27
New Home Sales	Negative	97	25.51***	0.77
Retail Sales	Positive	77	20.63**	0.66
Retail Sales	Negative	82	17.85**	0.55

Note: This table reports the performance of "Reversal" in accordance with recent macroeconomics news surprise over the period 1997 – 2020. Each strategy is implemented buying at the ask price and selling at the bid price, considering the closing value of each minute. "Events" indicates the number of FOMC meetings included. "Reversal" represents the average E-Mini S&P 500 return of the strategy across FOMC meetings, expressed in basis points. The Sharpe ratio is computed dividing the average return of the strategy (across events) by the standard deviation of the strategy returns and is annualized. Statistical significance of "Reversal" at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Table 9: Reversal in the (Quasi) Lucca and Moench Period.

Panel A – Different Strategies

Strategy	Holding Period	Events	Return	Sharpe Ratio
Reversal	From 1:50 p.m. $FOMC_t$ to 3:50 p.m. $FOMC_t$	109	14.30*	0.41
Drift	From 1:50 p.m. $FOMC_{t-1}$ to 1:50 p.m. $FOMC_t$	109	50.31***	1.05
Drift + Reversal	From 1:50 p.m. $FOMC_{t-1}$ to 3:50 p.m. $FOMC_t$	109	64.61***	1.17
Announcement Return	From the Ann. Minute to 3:50 p.m. $FOMC_t$	109	-6.68	-0.19
Open to Close $FOMC_t$	From 9:30 a.m. $FOMC_t$ to 3:50 p.m. $FOMC_t$	109	31.29***	0.80
Close to Close $FOMC_t$	From 3:50 p.m. $FOMC_{t-1}$ to 3:50 p.m. $FOMC_t$	109	33.07***	0.77

Panel B – Reversal Strategies

Reversal, Given:	Strategy	Events	Return	Sharpe Ratio
Negative Pre-Ann.	Buy at 1:50 p.m. and Sell at 3:50 p.m. $FOMC_t$	28	21.21*	0.72
Positive Pre-Ann.	Sell at 1:50 p.m. and Buy at 3:50 p.m. $FOMC_t$	81	11.90	0.33
Large Negative Pre-Ann.	Buy at 1:50 p.m. and Sell at 3:50 p.m. $FOMC_t$	14	39.55*	1.17
Large Positive Pre-Ann.	Sell at 1:50 p.m. and Buy at 3:50 p.m. $FOMC_t$	40	19.44	0.48
Small Negative Pre-Ann.	Buy at 1:50 p.m. and Sell at 3:50 p.m. $FOMC_t$	14	2.88	0.31
Small Positive Pre-Ann.	Sell at 1:50 p.m. and Buy at 3:50 p.m. $FOMC_t$	41	4.55	0.23

Note: This table reports the results of "Reversal" in the (Quasi) Lucca and Moench period, which spans from September 1997 to March 2011. "Reversal" is implemented as follows: if the prior 24-hour return at 1:50 p.m. $FOMC_t$ is negative (positive), buy (sell) at 1:50 p.m. and sell (buy) at 3:50 p.m. In Panel A, we compare "Reversal" with other possible strategies that an investor can implement around FOMC announcements, where the position is always opened with a buy order. In Panel B, we condition "Reversal" on the magnitude and direction of the prior 24-hour pre-announcement return (Pre-Ann.). Each strategy is implemented buying at the ask price and selling at the bid price, considering the closing value of each minute (except for the announcement minute, where the open value of the minute is used). "Events" indicates the number of FOMC meetings included. "Return" represents the average E-Mini S&P 500 return of the strategy across FOMC meetings, expressed in basis points. The Sharpe ratio is computed dividing the average return of the strategy (across events) by the standard deviation of the strategy returns and is annualized. Statistical significance of "Return" at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

Table 10: Reversal in the Post-Lucca and Moench Period.

Panel A – Different Strategies

Strategy	Holding Period	Events	Return	Sharpe Ratio
Reversal	From 1:50 p.m. $FOMC_t$ to 3:50 p.m. $FOMC_t$	71	23.34***	0.89
Drift	From 1:50 p.m. $FOMC_{t-1}$ to 1:50 p.m. $FOMC_t$	71	1.62	0.09
Drift + Reversal	From 1:50 p.m. $FOMC_{t-1}$ to 3:50 p.m. $FOMC_t$	71	24.95**	0.76
Announcement Return	From the Ann. Minute to 3:50 p.m. $FOMC_t$	71	-0.78	-0.03
Open to Close $FOMC_t$	From 9:30 a.m. $FOMC_t$ to 3:50 p.m. $FOMC_t$	71	-3.57	-0.13
Close to Close $FOMC_t$	From 3:50 p.m. $FOMC_{t-1}$ to 3:50 p.m. $FOMC_t$	71	6.72	0.22

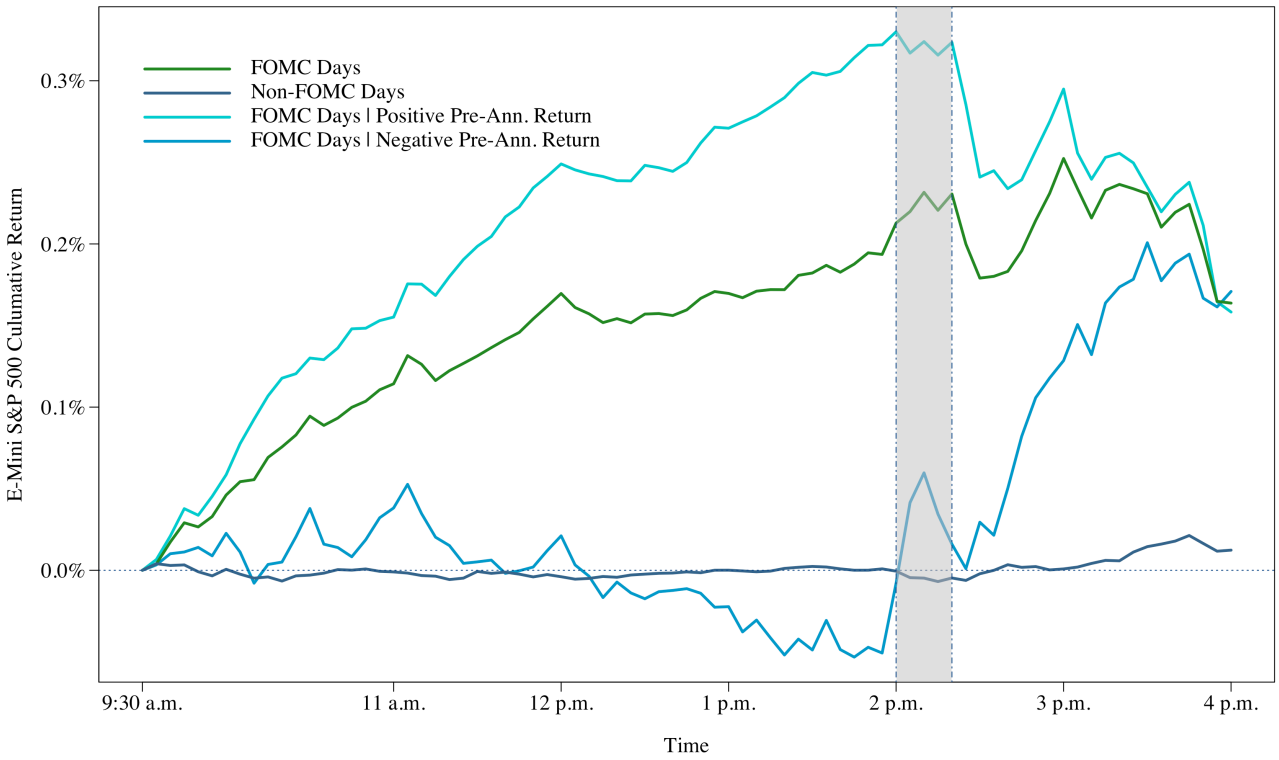
Panel B – Reversal Strategies

Reversal, Given:	Strategy	Events	Return	Sharpe Ratio
Negative Pre-Ann.	Buy at 1:50 p.m. and Sell at 3:50 p.m. $FOMC_t$	38	24.18**	0.85
Positive Pre-Ann.	Sell at 1:50 p.m. and Buy at 3:50 p.m. $FOMC_t$	33	22.37**	0.92
Large Negative Pre-Ann.	Buy at 1:50 p.m. and Sell at 3:50 p.m. $FOMC_t$	19	37.71**	1.19
Large Positive Pre-Ann.	Sell at 1:50 p.m. and Buy at 3:50 p.m. $FOMC_t$	16	24.90	0.84
Small Negative Pre-Ann.	Buy at 1:50 p.m. and Sell at 3:50 p.m. $FOMC_t$	19	10.64	1.14
Small Positive Pre-Ann.	Sell at 1:50 p.m. and Buy at 3:50 p.m. $FOMC_t$	17	19.98*	2.79

Note: This table reports the results of "Reversal" in the post-Lucca and Moench period, which spans from March 2011 to January 2020. "Reversal" is implemented as follows: if the prior 24-hour return at 1:50 p.m. $FOMC_t$ is negative (positive), buy (sell) at 1:50 p.m. and sell (buy) at 3:50 p.m. In Panel A, we compare "Reversal" with other possible strategies that an investor can implement around FOMC announcements, where the position is always opened with a buy order. In Panel B, we condition "Reversal" on the magnitude and direction of the prior 24-hour pre-announcement return (Pre-Ann.). Each strategy is implemented buying at the ask price and selling at the bid price, considering the closing value of each minute (except for the announcement minute, where the open value of the minute is used). "Events" indicates the number of FOMC meetings included. "Return" represents the average E-Mini S&P 500 return of the strategy across FOMC meetings, expressed in basis points. The Sharpe ratio is computed dividing the average return of the strategy (across events) by the standard deviation of the strategy returns and is annualized. Statistical significance of "Return" at the 1%, 5%, and 10% level is indicated by ***, **, and *, respectively.

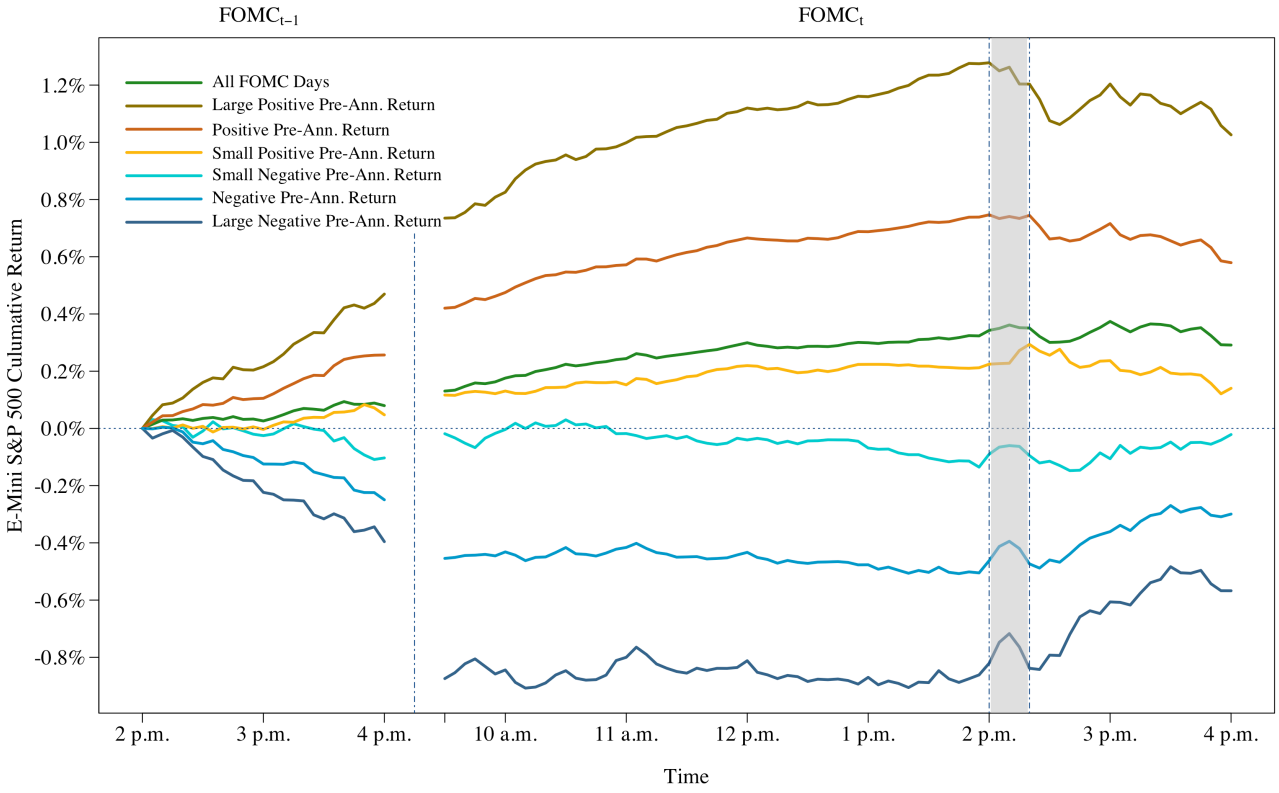
9 Figures

Figure 1: E-Mini S&P 500 Return on FOMC- and Non-FOMC Days.



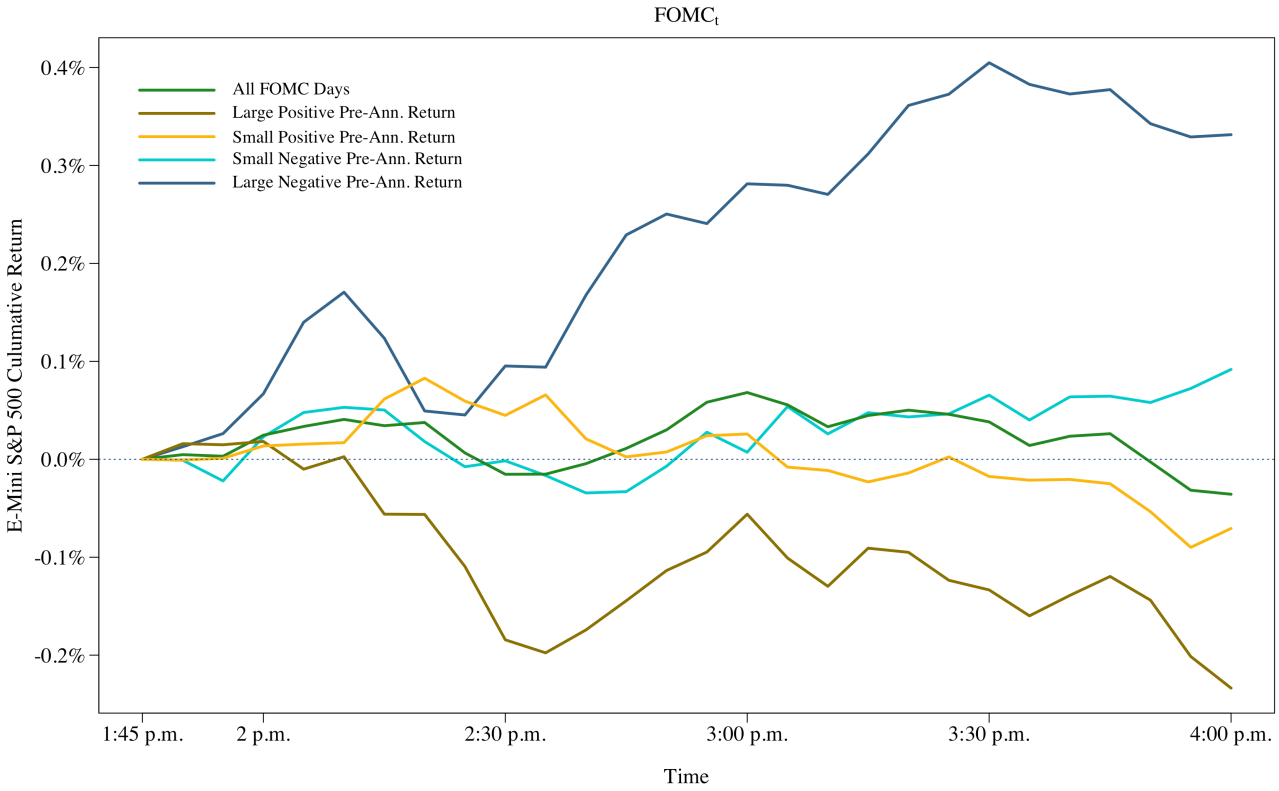
Note: This figure reports the average intraday E-Mini S&P 500 cumulative return across FOMC- and non-FOMC days from the opening time (9:30 a.m. EST) until the end of the trading day (4:00 p.m. EST), using one observation every 5 minutes. FOMC days are classified as characterized by a positive and a negative pre-announcement return in accordance to whether the 24-hour return ending 15 minutes before the announcement is positive or negative, over the full-sample period (09/1997 – 01/2020). Announcements time in the full-sample period is between 2:00 p.m. and 2:20 p.m. EST FOMC_t and is indicated by a gray-shaded area (we exclude the eight meetings when the policy statement was released at 12:30 p.m. EST).

Figure 2: Pre- and Post-Announcement Return Over the Period 1997 – 2020.



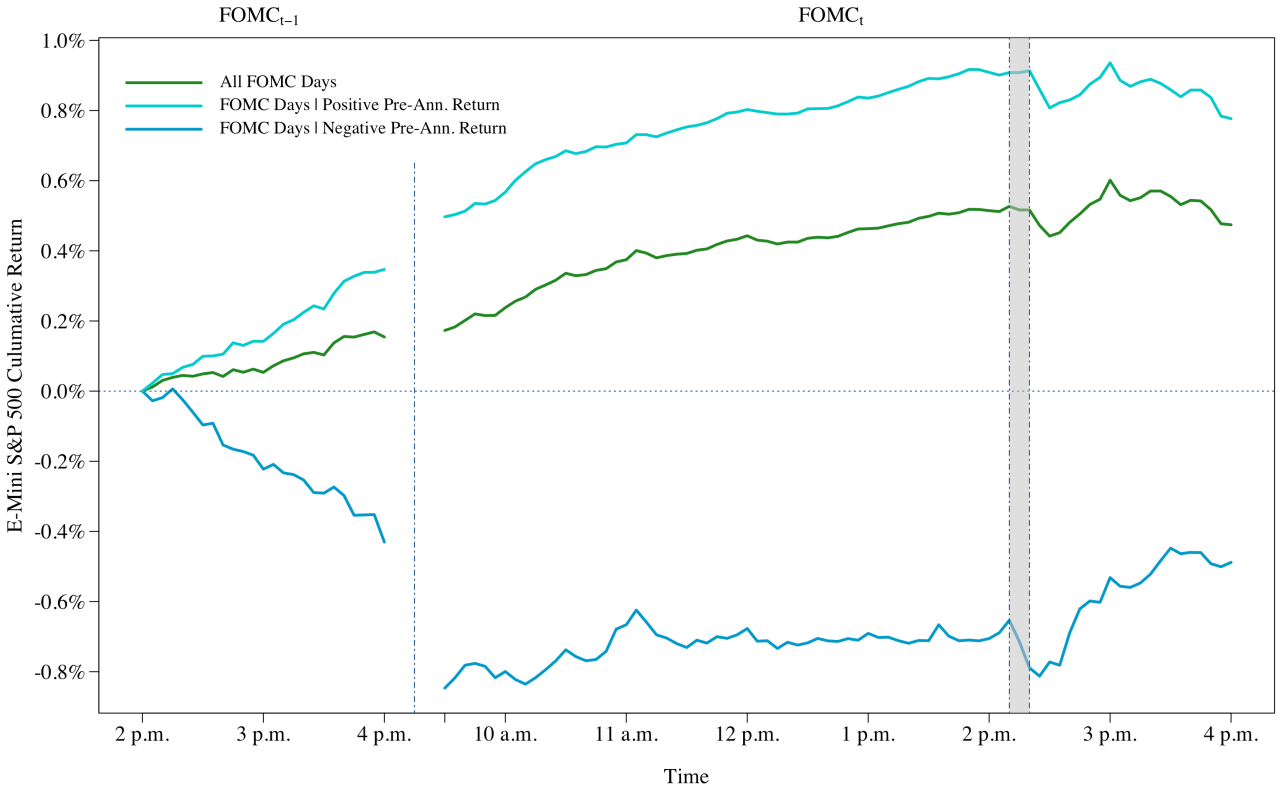
Note: This figure reports the average intraday E-Mini S&P 500 cumulative return across FOMC meetings from 2:00 p.m. EST FOMC$_{t-1}$ until the closing time of FOMC$_t$, using one observation every 5 minutes. We classify announcements according to the magnitude and direction of the 24-hour return ending 15 minutes before the announcement, over the full-sample period (09/1997 – 01/2020). Announcements time in the full-sample period is between 2:00 p.m. and 2:20 p.m. EST FOMC$_t$ and is indicated by a gray-shaded area (we exclude the eight meetings when the policy statement was released at 12:30 p.m. EST).

Figure 3: E-Mini S&P 500 Return Given the Magnitude of the Pre-Announcement Return.



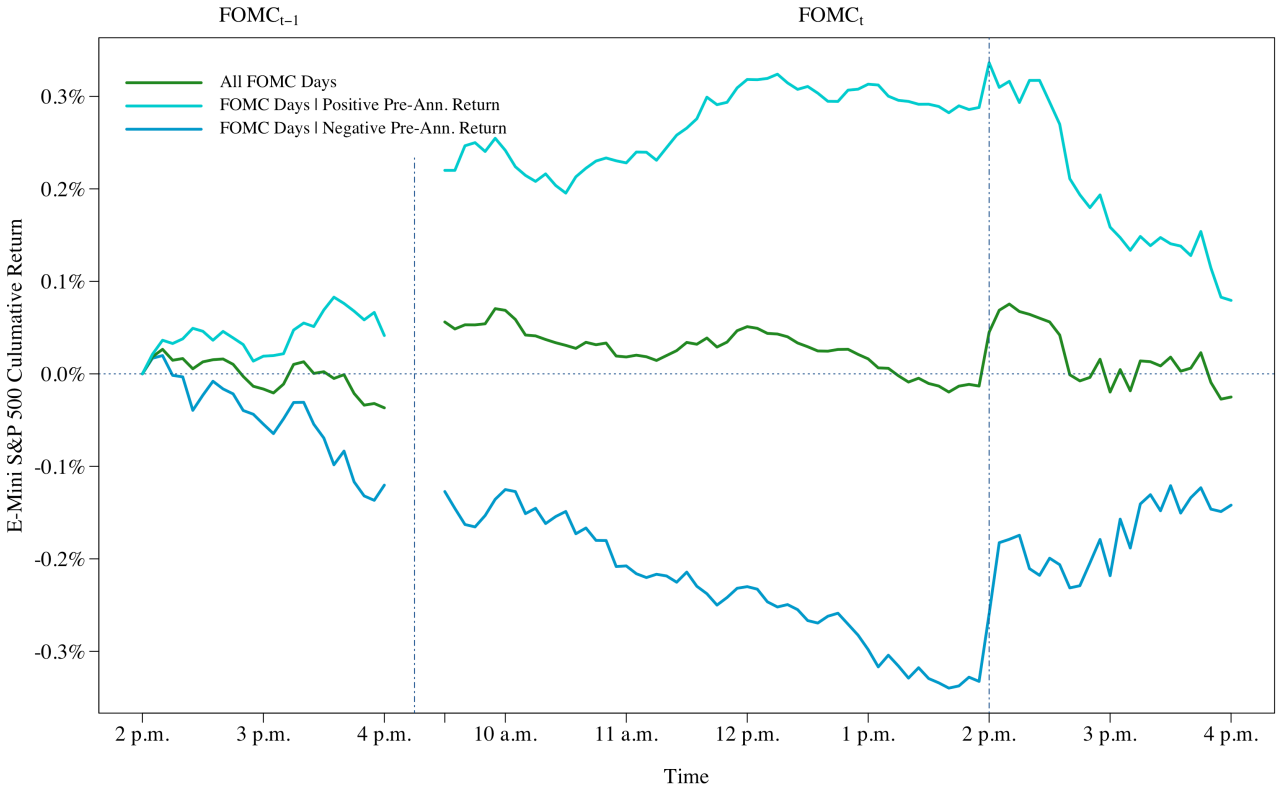
Note: This figure reports the average E-Mini S&P 500 cumulative return across FOMC meetings starting at 1:45 p.m. EST on FOMC$_t$, using one observation every 5 minutes. We classify announcements according to the magnitude and direction of the 24-hour return ending 15 minutes before the announcement, over the full-sample period (09/1997 – 01/2020). Announcements time in the full-sample period is between 2:00 p.m. and 2:20 p.m. EST FOMC$_t$ (we exclude the eight meetings when the policy statement was released at 12:30 p.m. EST).

Figure 4: Pre- and Post-Announcement Return Over the Period 1997 – 2011.



Note: This figure reports the average intraday E-Mini S&P 500 cumulative return across FOMC meetings from 2:00 p.m. EST FOMC$_{t-1}$ until the closing time of FOMC$_t$, using one observation every 5 minutes. We classify announcements according to the direction of the 24-hour return ending 15 minutes before the announcement, over the (Quasi) Lucca and Moench period (09/1997 – 03/2011). Announcements time in the (Quasi) Lucca and Moench period is between 2:11 p.m. and 2:20 p.m. EST FOMC$_t$ and is indicated by a gray-shaded area.

Figure 5: Pre- and Post-Announcement Return Over the Period 2011 – 2020.



Note: This figure reports the average intraday E-Mini S&P 500 cumulative return across FOMC meetings from 2:00 p.m. EST FOMC$_{t-1}$ until the closing time of FOMC$_t$, using one observation every 5 minutes. We classify announcements according to the direction of the 24-hour return ending 15 minutes before the announcement, over the post-Lucca and Moench period (04/2011 – 01/2020). Announcements time in the post-Lucca and Moench period is at 2:00 p.m. EST FOMC$_t$ (we exclude the eight meetings when the policy statement was released at 12:30 p.m. EST).